Introduction to Programming

Lecture 13

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Today's Lecture

- Manipulation of Two dimensional arrays
- Analyzing and solving a real world problem

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Array Manipulation

Example 1

Input

Row 1	1	2	3
Row 2	4	5	6
Row 3	7	8	9

Output

Memory

Row 3	7	8	9
Row 2	4	5	6
Row 1	1	2	3

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Addressing Array Elements

a [rowIndex] [columnIndex]

Example 1

```
int row ;
int col ;
const maxRows = 3 ;
const maxCols = 3 ;
int a [ maxRows ] [ maxCols ] ;
```

Example 1

```
for ( row = 0 ; row < maxRows ; row ++ )
{
 for ( col = 0 ; col < maxCols ; col ++ )
 {
  cout << "Please enter value of     element number "<<row<<
   "," << col ;
  cin >> a [ row ] [ col ] ;
 }
}
```

Example 2

maxRows = 3 ;

maxCols = 3 ;

Index of Start

Index of Last Row = maxRows - 1

[0]	1	2	3
[1]			
[2]			

Example 2

```
for ( row = maxRows - 1 ; row >= 0 ; row -- )
{
 for ( col = 0 ; col < maxCols ; col ++ )
 ...
}
```

Decrement Operator

Row 1	1	2	3
Row 2	4	5	6
Row 3	7	8	9

Row 3	7	8	9
Row 2	4	5	6
Row 1	1	2	3

Example 2: Formatted Output

```
cout << "The original matrix is" ;
for ( row = 0 ; row < maxRows ; row ++ )
{
 for ( col = 0 ; col < maxCols ; col ++ )
 {
  cout <<  a [ row ] [ col ]  ;
 }
}
```

```
<< '\t' ;
```

```
15 42
```

Example 2: Formatted Output

```
for ( row = 0 ; row < maxRows ; row ++ )
{
  for ( col = 0 ; col < maxCols ; col ++ )
  {
   cout << a [ row ] [ col ] << '\t' ;
  }
 cout << ' \n ' ;
}
```

```
15    42    26    7
```

Exercise

Enter the values in a matrix and print it in reverse Column order

[0]	[1]	[2]
1	2	3
4	5	6
7	8	9

[2]	[1]	[0]
3	2	1
6	5	4
9	8	7

Transpose of a Matrix

1	2	3
4	5	6
7	8	9

Square Matrix

Number of rows are equal to number of columns

$$\text{arraySize} = \begin{cases} \text{rows} \\ \text{cols} \end{cases}$$

Square Matrix

$$a_{ij} = a_{ji}$$

i = rows

j = columns

Square Matrix

```
int a [ row ] [ col ] ;
int arraySize ;
for ( row = 0 ; row < arraySize ; row ++ )
{
 for ( col = 0 ; col < arraySize ; col ++ )
 {
  //Swap values
 }
}
```

Swap Mechanisms

```
temp = a [ row ] [ col ] ;
a [ row ] [ col ] = a [ col ] [ row ] ;
a [ col ] [ row ] = temp ;
```

Practical Problem

Problem statement

Given tax brackets and given employee gross salaries , determine those employees who actually get less take home salary than others with lower initial income

Rule for tax deduction

0	–> 5,000	No tax
5001	– >10,000	5% Income Tax
10,001	– >20,000	10% Income Tax
20,001 and more		15% Income tax

Example

Net salary = Rs 10,000

Tax = **5%**

Amount Deducted = 5% of 10,000

= **500**

Net amount after deduction = 10,000 - 500

= **9,500**

Net salary = Rs 10,001

Tax = **10%**

Amount Deducted = 10% of 10,001

= **1,000.1**

Net amount after deduction = 10,001 - 1,000.1

= **9,000.9**

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Storage Requirement

One- dim arrays of integer

lucky = 0

lucky = 1

Storage of salary

No of Emp.	Grow Salary	Net Salary After Deduction
1	5,000	5,000
2	10,000	9,500
3		
4		
5		
6		
7		
8		
9		
10		

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Interface Requirements

Distribution of the Program

- Input
- Salary calculation
- Identification of the unlucky individuals
- Output

Detail Design

Functions in the program

getInput
calculateSalary
locateUnluckyIndividual
displayOutput

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Code

```
#include<iostream.h>

void getinput ( int [ ] [ 2 ] , int ) ;
main ( )
{
 const int arraySize = 100 ;
 int sal [ arraySize ] [ 2 ] ;
 int lucky [ arraySize ] = { 0 } ;
 int numEmps ;
 cout << "Enter the number of employess " ;
 cin >> numEmps  ;
 getInput ( sal , numEmps ) ;
}
```

Code

```
getInput ( int sal [ ] [2] , int numEmps )
{
 for ( i = 0 ; i < numEmps ; i ++ )
  cin >> sal [ i ] [ 0 ] ;
}
```

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Exercise

Suppose you are given a square matrix of size n x n , write a program to determine if this is an identity matrix